WORKSHEET · PYTHON (GR 6–8) BEGINNER: PYTHON GAME CODERS

Lesson 8: Capstone: Treasure Hunt

Module 2 | 60 minutes | game design, lists, def | Grades 6–10 · ages 11–15

Name: Class: Date:

Plan your project and practise the key ideas.

1. Key words

Match each word to what it means. Write the letter.

a. text adventure ___ A. The word that defines a new function.
b. inventory ___ B. A named block of code you can run again and again.
c. function ___ C. A list of items the player is carrying.
d. def ___ D. A game where the player types commands to explore a story.

2. Plan your program

Before you type it: Your first function. Write def describe_room(name): that prints a border line and then `You are in the ...` with the room name. Call it three times, for the Hall, the Kitchen and the Cave. Write the steps in plain English (pseudocode), then the Python you will use.

3. Predict the output: what is in the bag?

What does this program print? Code: bag = [] ↵ bag.append("torch") ↵ bag.append("key") ↵ if "map" in bag: ↵ print("You know the way!") ↵ else: ↵ print("You have", len(bag), "things.")

a. A. You have 2 things.
b. B. You know the way!
c. C. You have 3 things.
d. D. Nothing

4. Design a game

Put the game designer’s steps in order Number them 1–6.

a. ___ Add the other rooms and test each one
b. ___ Code the game loop and one room
c. ___ Write the win and lose rules
d. ___ Choose where the key and treasure go
e. ___ Draw a map of the rooms
f. ___ Playtest with a friend and improve

5. Plan your program

Before you type it: Debug it! This adventure start-up has three bugs. Fix it so it prints `Welcome, Explorer!` and then `You can see in the dark.` Write the steps in plain English (pseudocode), then the Python you will use.

6. My plan

Plan your project: Design and build a text adventure with at least three rooms, an inventory, a locked door, a treasure to win and a torch that runs out. Sketch it and list what each part will do.

7. Checklist

Tick each one when your project does it.

a. ☐ A paper map of 3+ rooms, copied into a comment
b. ☐ An inventory list that grows with append and is checked with in
c. ☐ A while loop game with if / elif for every room and command
d. ☐ A win ending and a lose ending (GAME OVER after 10 moves)
e. ☐ At least one function you define with def and call
f. ☐ quit ends the game and unknown commands do not crash it

8. Two stars and a wish

Ask a friend to test your project. Write two things they liked and one idea to improve it.

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ I can plan a game with a map, a win rule and a lose rule.
☐ I can use a while loop, a room variable and if / elif to run a text adventure.
☐ I can use an inventory list with append and in.
☐ I can define and call a simple function with def.